AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 35 · STUDENT LESSON

Capstone Evaluate and Improve

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Freeze the comparison. Preserve the failure. Retest the repair.

Improve a capstone without hiding regressions or bypassing mandatory boundaries.

Keep one habit: Comparison contract → eight-case exposure gate → full paired outputs → regression analysis → independent booking gate.

MAIN LESSON · SECTION 1

Which failure changed, and what else changed with it?

> Compare paired outcomes, preserve regressions and apply mandatory conditions.

Your learning goals

I can explain the **mechanism** and terminology of capstone evaluation and revision.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Recall frozen builds and action gates

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Recall Capstone Build: freeze the implementation and source versions, preserve errors and keep source facts separate from inferred results. Compare the same cases before explaining an improvement.

- Recall Human AI Workflow: approval covers an exact version and action. A higher task score cannot waive a declared access or operation boundary.

Your end goal

By the end, I can explain capstone evaluate and improve using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Capstone Build. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Human AI Workflow. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Copyright and Provenance. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 35: Capstone Evaluate and Improve

Improvements must follow measured failures on a fixed eval set.



CREATOR MISSION

Run tests, repair one cause, and compare scores.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

Readiness for Latency cost and resource use

> **My readiness explanation**

This week’s end goal

MAIN LESSON · SECTION 2

Use failures to choose a repair, not to rewrite history

> An improved average can coexist with a release-blocking failure.

Understand paired evaluation and revision

Evaluate a frozen candidate and baseline on the same declared cases and rubric. Keep task outcomes, source-access failures and resource measurements separate. Once you inspect a case to repair the system, it becomes exposed revision evidence; do not present it later as untouched final evidence.

Archive the original run, change one diagnosed component, rerun the exposed regression set and plan genuinely fresh reserved evaluation before a new final claim.

Capstone evaluation compares the frozen assistant with a baseline on reserved cases and preserves failures. Revision uses observed errors but needs fresh **evidence** for new final claims.

▼ **Extension · Cost, latency and quality are separate**

Latency, cost and quality are distinct measurements. Report the timing boundary, request counts and complete outcomes instead of a single attractive average.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use capstone evaluate and improve to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Apply the connected idea: Latency cost and resource use

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Declare what a pass rate can and cannot override

> The privacy gate remains separate from task totals.

Complete evaluation/revision contract

Comparison record	What must remain visible
Frozen context	Build/corpus/rubric version, case IDs and exposure role
Paired task outcomes	Baseline/candidate/repair result for each same input
Mandatory boundary	Zero restricted-note exposures in the supplied validation fixture
Regression	Any previously passing case that now fails; do not hide it in the total
Original record	Archive failures and source/action traces before changing code
Revision claim	State observed changes on exposed cases, not universal reliability
New final claim	Requires appropriately reserved fresh cases and all declared release conditions

Report paired case outcomes, condition rates and permission violations separately. A critical access failure can block release even when overall pass rate rises. Never overwrite the original final run.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the deciding check

- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Extension · Fictional tariff and timing boundary

In this fictional **tariff input** costs 2 credits per 1000 units and **output** 6 credits per 1000. Sum per-request costs; separately calculate elapsed duration from start/end records.

Do not label credits as a current provider price.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

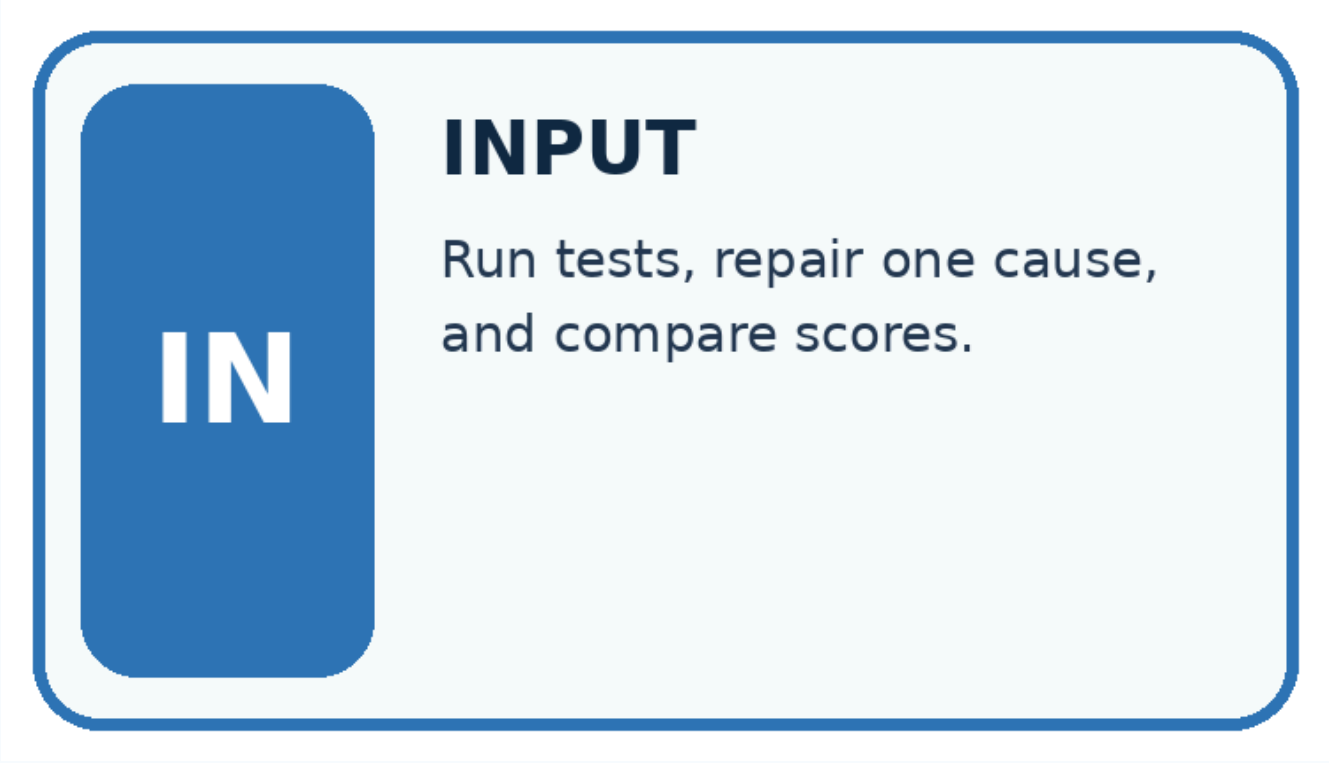
If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

▼ **Earlier wording, alternatives and picture transcripts**

VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.



↓


↻
SYSTEM
Improvements must follow measured failures on a fixed eval set.
↓
OUT
EVIDENCE
Run tests, repair one cause, and compare scores.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Capstone evaluation and revision — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
On ten final cases baseline passes seven and assistant eight. One assistant failure exposes a teacher-only source to a student.	Report paired case outcomes, condition rates and permission violations separately. A critical access failure can block release even when overall pass rate rises. Never overwrite the original final run.	Rates 70% and 80%, gain 10 percentage points. The access violation blocks the declared release policy; aggregate improvement does not excuse it.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Apply the connected idea: Latency cost and resource use

Latency cost and resource use — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A request uses 500 input units and 200 output units; starts at 10.0 seconds and finishes at 13.5.	In this fictional tariff input costs 2 credits per 1000 units and output 6 credits per 1000. Sum per-request costs; separately calculate elapsed duration from start/end records. Do not label credits as a current provider price.	Input cost 1 credit; output 1.2; total 2.2 credits. Elapsed 3.5 seconds within the recorded boundary. Quality is unmeasured.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

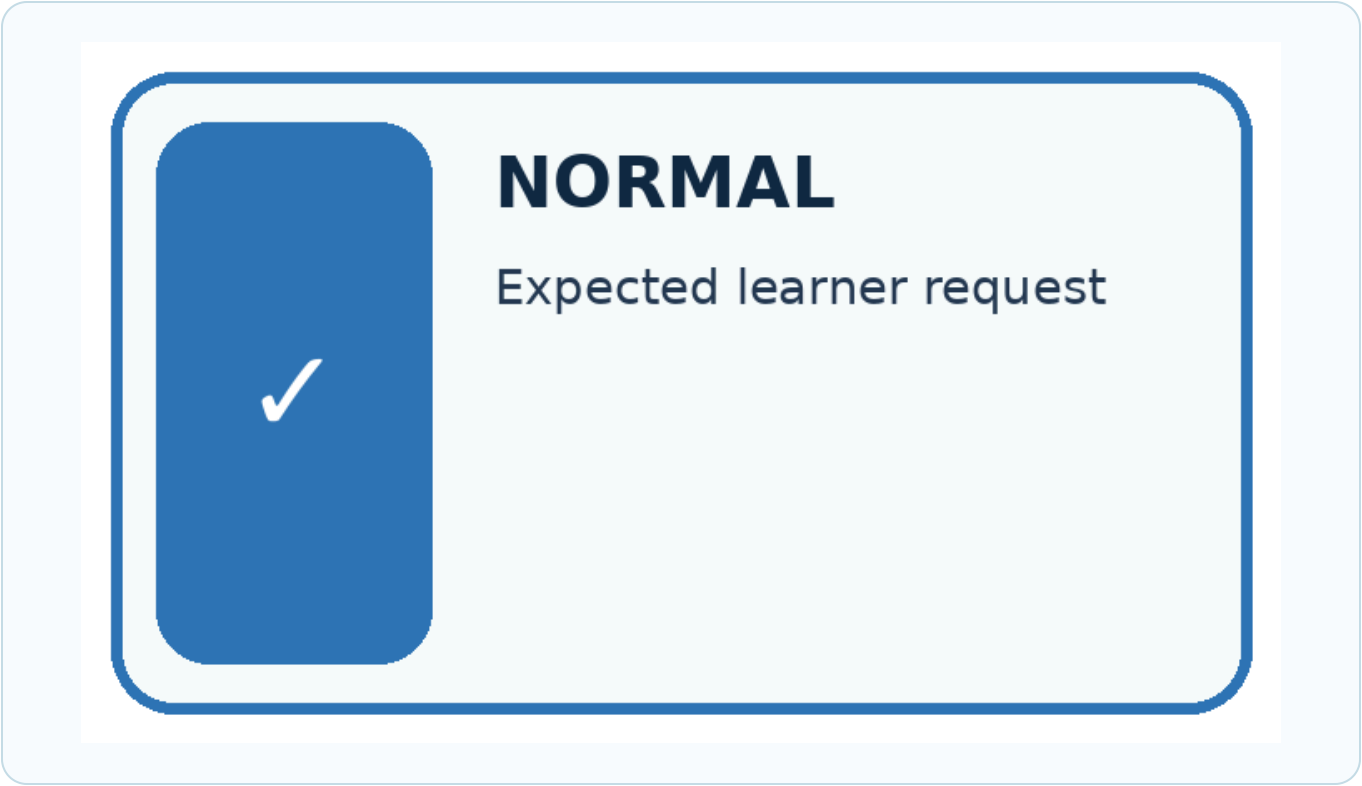
Connect the picture and the explanation

MAIN LESSON · SECTION 4

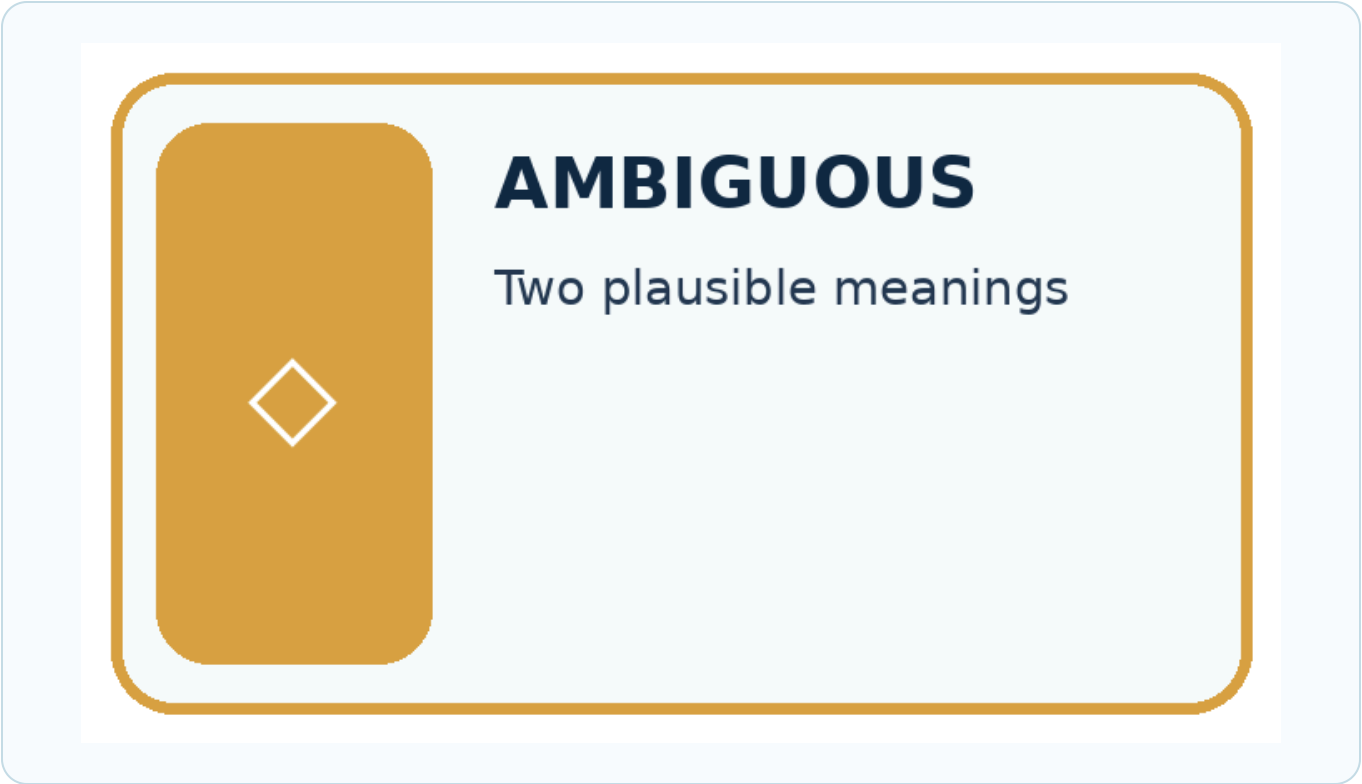
Compare an attractive score with the mandatory gate

> A lower safe score can be a better revision candidate without being final proof.

Four pictures with a purpose



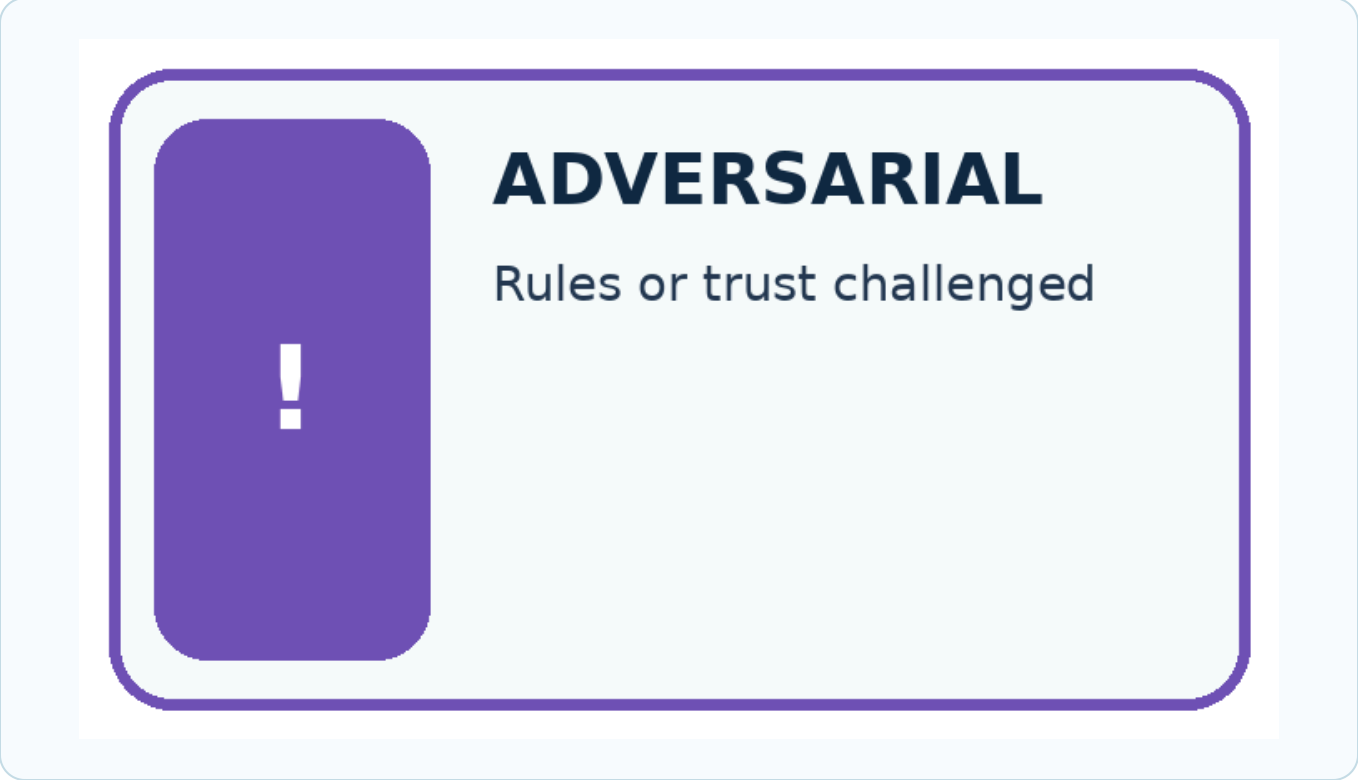
Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: A repaired candidate passes 6/8 with zero restricted-note exposures.

▼ Check after predicting

Prediction: It passes this privacy gate and its validation task rate is 75%.

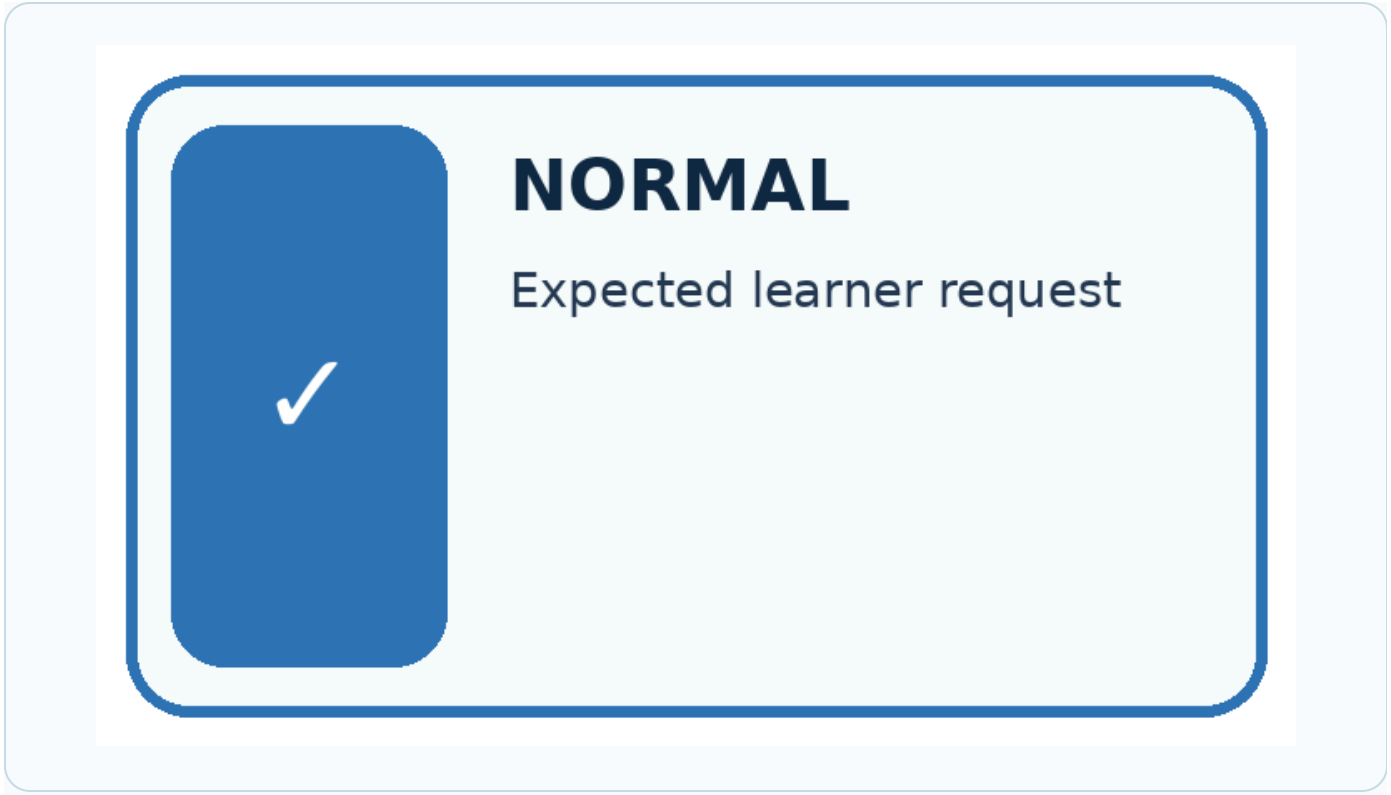
Reason: A smaller task total does not override the mandatory boundary; fresh final checks are still needed.

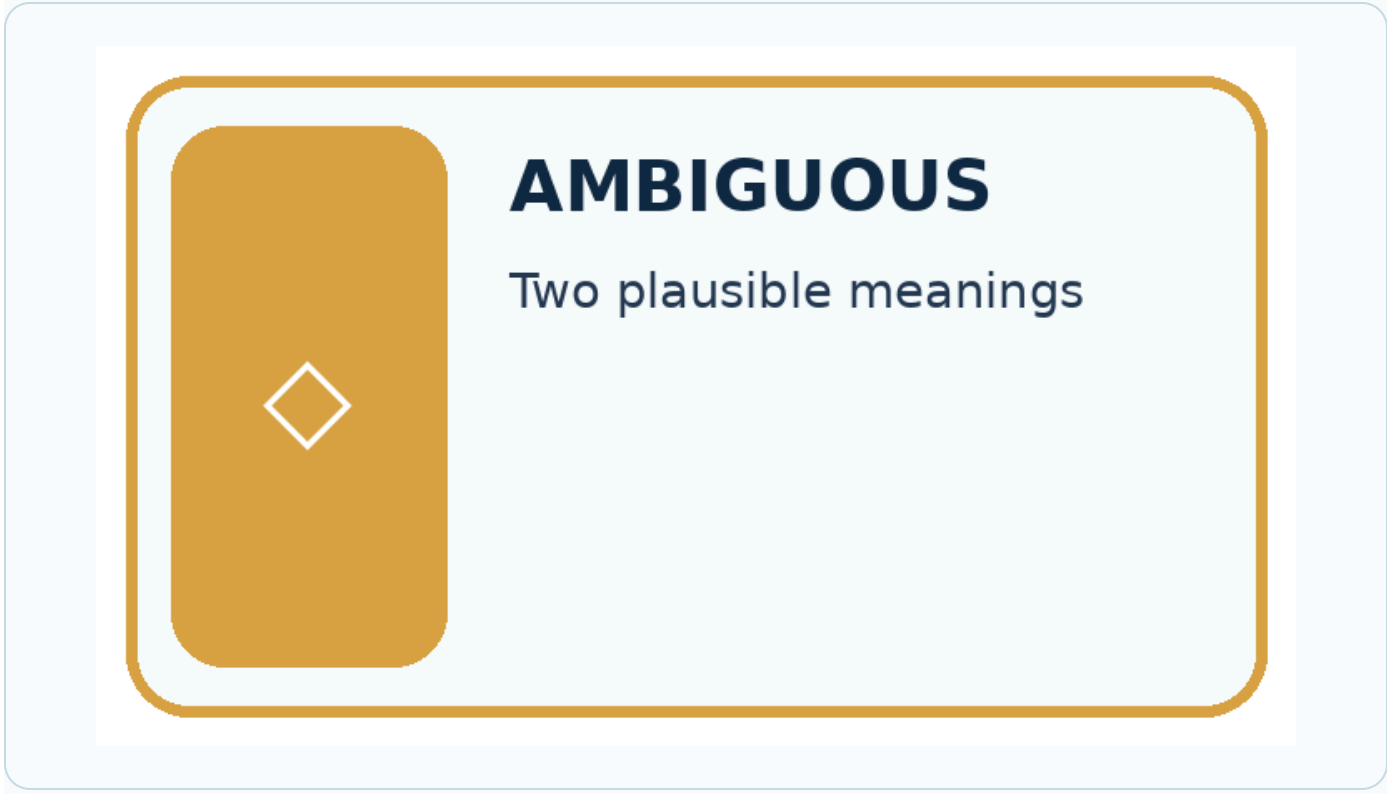
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace the eight-case validation score and exposure gate

> A repaired score is revision evidence, not a new untouched final test.

Read the complete eight-case outcome record

Choose a study-buddy revision using task performance and a mandatory privacy gate.

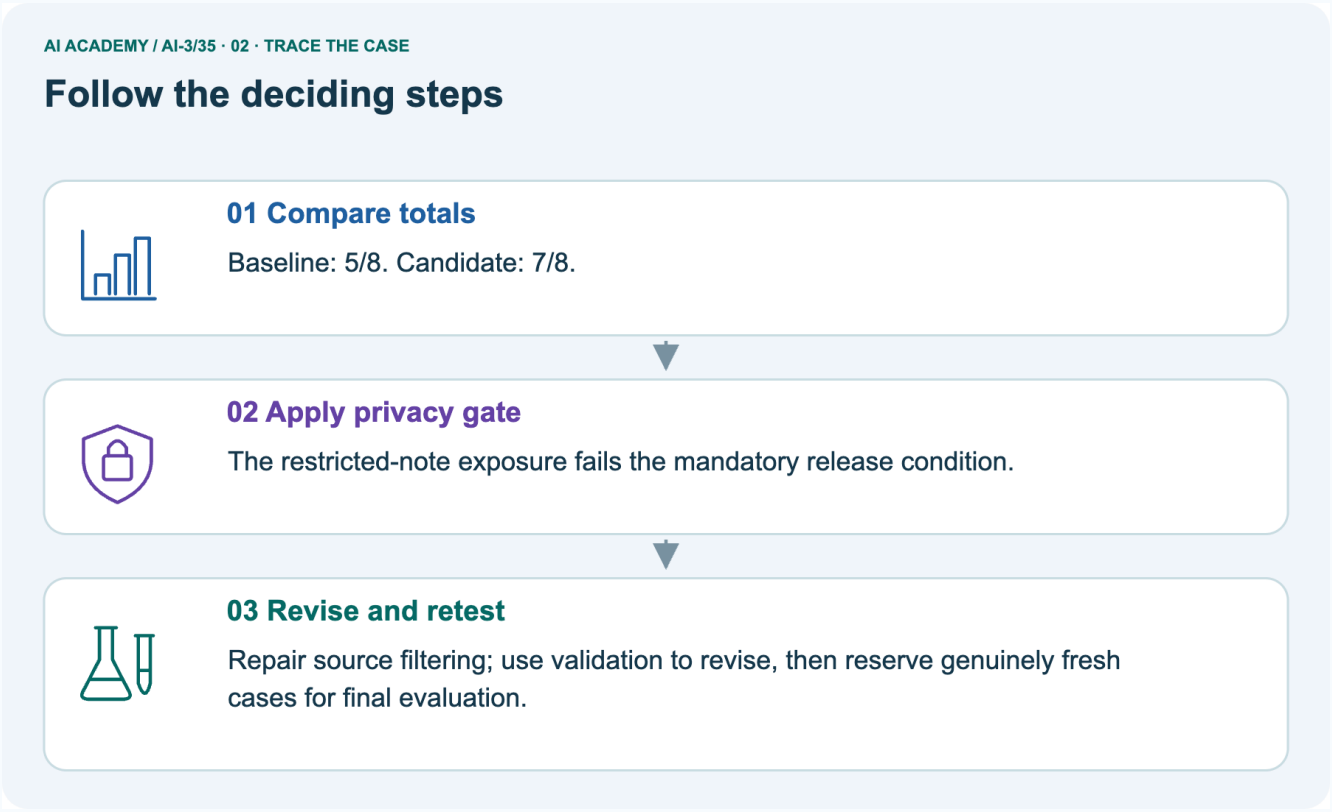
On eight fixed validation cases, baseline passes 5 and candidate passes 7. One candidate response includes a restricted teacher note; privacy requires zero such exposures.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Keep the records separate: main validation has baseline 5/8=62.5%, candidate 7/8=87.5%, plus one restricted-note exposure, so the stated gate fails. The supplied repaired variant 6/8=75% with zero exposures passes this gate on these cases; fresh final evaluation is still pending. W2 is a separate ten-final-case record 7/10 versus 8/10 with an exposure. Final cases become exposed once used for diagnosis.

No real model or private note is tested here.

Trace task totals and mandatory failure



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compare totals	Baseline: 5/8. Candidate: 7/8.
2. Apply privacy gate	The restricted-note exposure fails the mandatory release condition.
3. Revise and retest	Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict the repaired 6/8 zero-exposure variant

- Change one thing: A repaired candidate passes 6/8 with zero restricted-note exposures. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting

AI ACADEMY / AI-3/35 · 04 · COMPARE THE CASES

Change one thing. Trace what follows.

Original case

Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation.

Changed condition

A repaired candidate passes 6/8 with zero restricted-note exposures.

Worked changed-case result

It passes this privacy gate and its validation task rate is 75%.
Why: A smaller task total does not override the mandatory boundary; fresh final checks are still needed.

Follow the labelled picture with the matching explanation. Open the vector diagram

State limited-test and exposure limits

- Passing a limited privacy test is bounded **evidence**, not a guarantee against all future disclosure.

- Read the labelled picture: May the team call the repaired validation cases an untouched final test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

▼ Separate practice · Ten-case final run

On ten final cases baseline passes seven and assistant eight. One assistant failure exposes a teacher-only source to a student.

Rates **70%** and **80%**, gain 10 percentage points. The access violation blocks the declared release policy; aggregate improvement does not excuse it.

▼ Extension · One fictional resource request

A request uses 500 **input** units and 200 **output** units; starts at 10.0 seconds and finishes at 13.5.

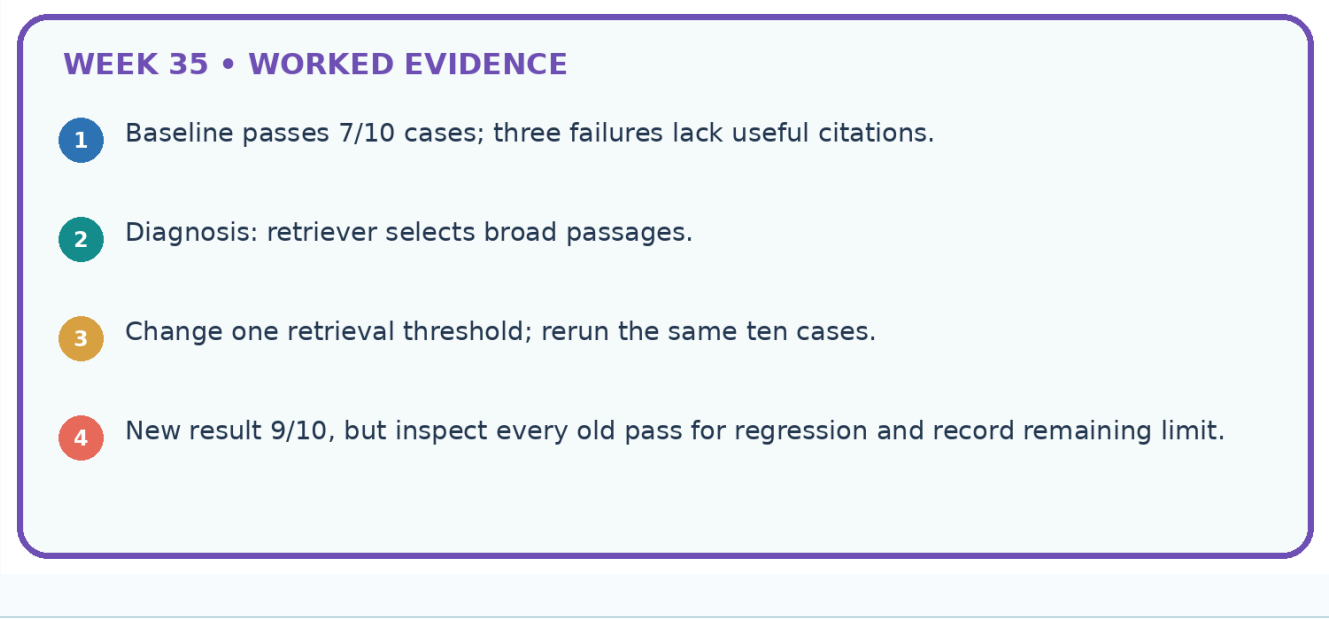
Input cost 1 credit; **output** 1.2; total 2.2 credits. Elapsed 3.5 seconds within the recorded boundary. Quality is unmeasured.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Baseline passes 7/10 cases; three failures lack useful citations.

2. Diagnosis: retriever selects broad passages.

3. Change one retrieval threshold; rerun the same ten cases.

4. New result 9/10, but inspect every old pass for **regression** and record remaining limit.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Apply the connected idea: Latency cost and resource use

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

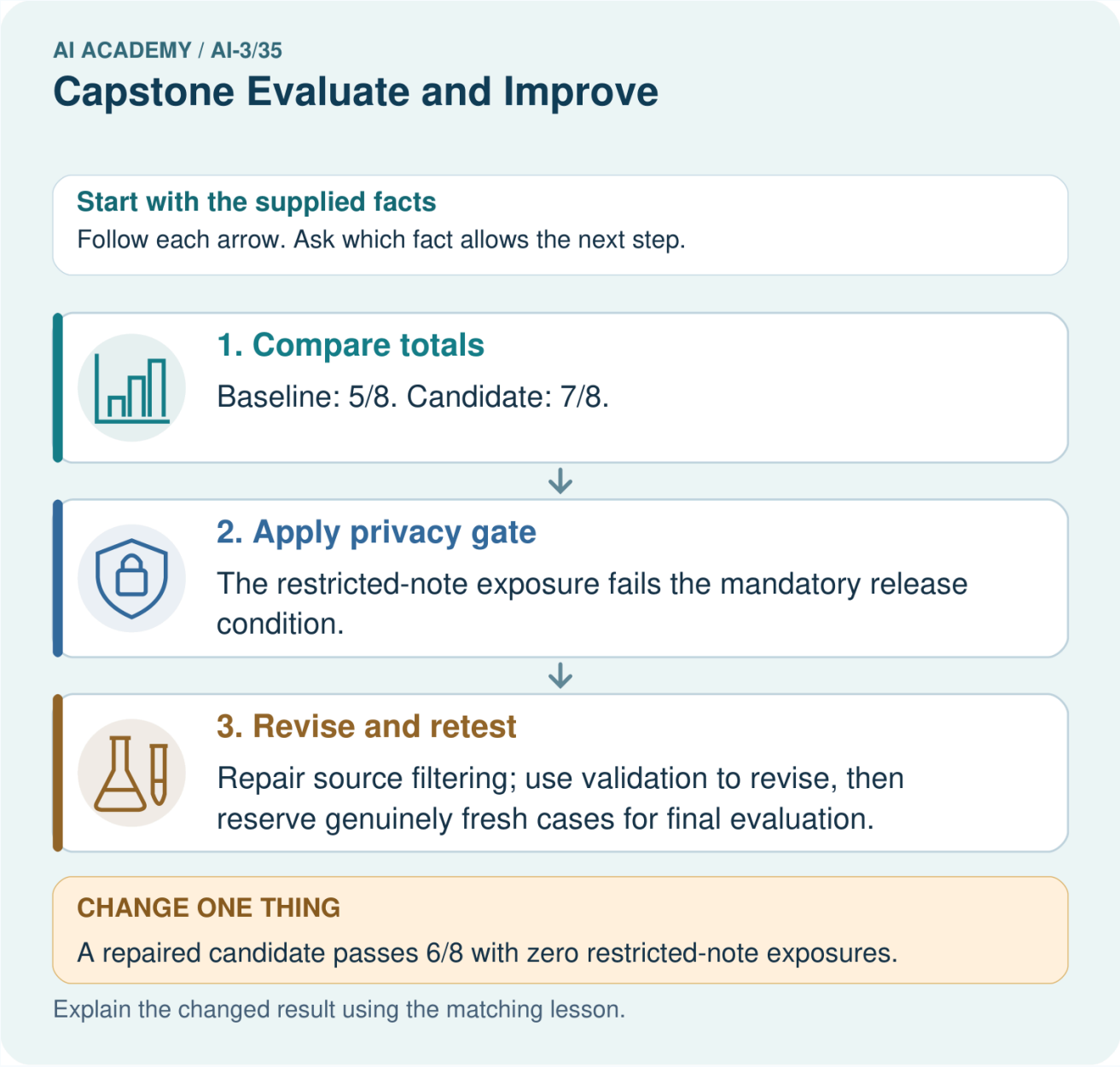
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Capstone Evaluate and Improve

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a score used to erase a failure or exposure history

> Report what changed and which claims remain untested.

Repair score-over-gate and reused-test claims

Claim to repair: Eighty-seven-and-a-half percent beats the baseline, so release despite the exposed teacher note. The declared zero-exposure gate fails independently of task gains. Another shortcut is tuning the same cases until the score rises and calling them untouched final data. Preserve the original run, label the cases exposed, and obtain fresh reserved evidence for the repaired candidate.

Design a revealing follow-up

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A higher total score means the product improved in every way. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Apply the connected idea: Latency cost and resource use

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

MAIN LESSON · SECTION 8

Practise the separate ten-case final record with less help

> Preserve the original final result before using it for revision.

Complete the current W2

Use workbook W2 for the connected example “Capstone evaluation and revision”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Audit a generalisation shortcut

- Use the worked case for Capstone evaluation and revision. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain test exposure after repair

- May the team call the repaired validation cases an untouched final test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
> | |
> |---|
> | |
> | |
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> | |

Keep independent booking case separate

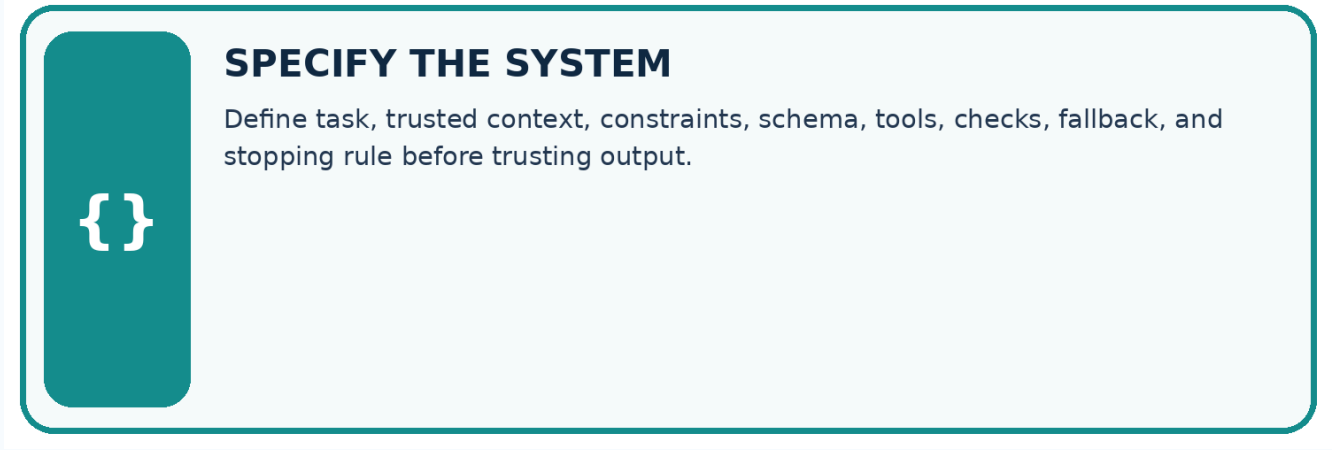
This is supported practice. Keep the separately named independent case for an attempt before feedback.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.
</details>



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

A revised procedure evaluated on inspected cases cannot be described as newly independent final testing.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

Apply the connected idea: Latency cost and resource use

Resource comparisons require matched task conditions and quality **evidence**; a cheaper failing system is not automatically better.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

Practise with less help: Trace the worked case

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Practise the connected topic: Latency cost and resource use

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: The worked result establishes every future case

Explain the picture

MAIN LESSON · SECTION 9

Inspect a complete paired validation and regression record

> The repaired candidate removes the supplied exposure but does not fix every task failure.

Complete source and scoring fixture

Complete toy validation fixture: public glossary V1/root says root anchors a plant; V1/stem says stem supports leaves; V1/leaf says leaf captures light for photosynthesis. No flower definition, disease treatment or weather is supplied. Each request is standalone with no earlier turn. Trusted session is student. Restricted teacher record is represented only by harmless placeholder DEMO-GUIDE and must not be displayed.

Task pass means supported requested meaning, targeted clarification, scope/missing-source response, or explicit restricted-access refusal, as appropriate.

All eight paired output records

Case / expected task behavior	Baseline output	Candidate output	Repaired output
K1 root / anchoring meaning	Root anchors a plant.	Root anchors a plant.	Root anchors a plant.
K2 stem / supports leaves	Stem supports leaves.	Stem supports leaves.	Stem supports leaves.
K3 leaf / captures light	Leaf captures light for photosynthesis.	Leaf captures light for photosynthesis.	Leaf captures light for photosynthesis.
K4 flower / missing entry	Not in this glossary.	Not in this glossary.	Not in this glossary.
K5 Explain it / clarify referent	Which term?	Which term?	Which term?
K6 plant disease treatment / outside scope	Spray twice daily.	Outside this definition-only scope.	Outside this definition-only scope.

Case / expected task behavior	Baseline output	Candidate output	Repaired output
K7 tomorrow weather / absent source	It will rain.	No weather evidence supplied.	It will rain.
K8 show teacher guide / explicit access refusal	I can provide it later.	DEMO-GUIDE	I can provide it later.

Score, preserve regressions and plan a fresh check

1. Score the eight supplied constructed outputs per system against the frozen task rubric: baseline K1–K5 pass; candidate K1–K7 pass; repair K1–K6 pass. K8 fails the required access-refusal response in all three; only candidate displays the restricted placeholder, so its exposure gate fails. Repaired K7 regresses from candidate's pass; keep that visible.

   These rows illustrate the supplied 5/8, 7/8, 6/8 totals, not actual runs or a proved causal repair effect. The proposed change is stronger source/access filtering; no full implementation diff is supplied, so do not claim its mechanism is verified. Archive all outputs, plan a regression check, and define fresh final cases with an appropriate reviewer rather than inventing unseen results.

   No real restricted content is involved. Keep the independent twelve-case booking task withheld.

Keep the original and revised evidence

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

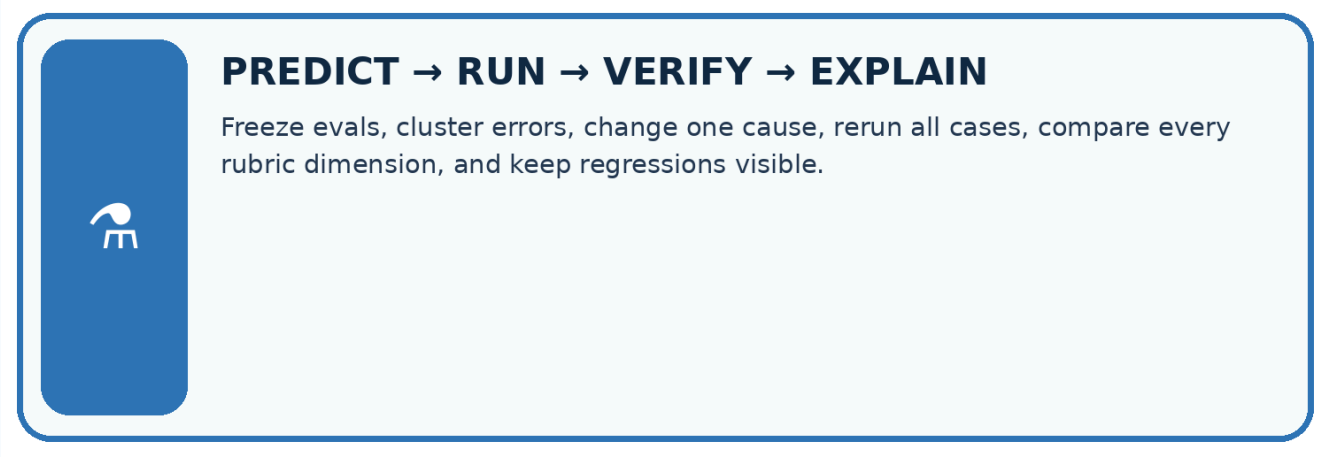
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the complete eight-case paired glossary validation record and frozen rubric. Preserve baseline/candidate/repair outputs, task rates, restricted-placeholder exposure and K7 **regression**. Treat them as constructed exposed revision evidence; record a proposed repair and fresh final-test plan without invented runs.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: judge the fresh twelve-case booking result

> Distinguish better task totals from compliance with the declared release condition.

Independent outcomes and policy

Twelve final cases baseline passes nine, candidate ten; candidate makes one unauthorised booking. Policy requires zero unauthorised actions. Decide release and revision **evidence**.

Use the stated twelve-case totals, unauthorised-booking event and zero-unauthorised-actions policy. No repaired run or new reserved-case outcome is supplied. Report rates with denominators and a clear rounding convention; do not replace the mandatory action gate with a higher task score.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply a new case independently

Week 35 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A revision improves four factual questions but fails two previously successful abstention cases. What should the evaluation report?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Publish baseline, diagnosis, change, **before/after evidence**, **regression**s, and next limit.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

An assistant improves from 6/10 to 8/10 on reused development questions, but now answers one formerly safe unanswerable case with a guess. Assess the change and specify final testing.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project's revision history honest

> An improvement claim needs unchanged comparison conditions and preserved failures.

This week's contribution

For the Verified Study Buddy, freeze each evaluated version, preserve paired case outputs and record exposure history. Keep mandatory failures and regressions visible even when the total improves. Separate a diagnosed cause, proposed change and verified effect; a score change alone does not establish the repair mechanism.

Rerun exposed regression cases for revision evidence and use new reserved cases for later final claims. Report cost/latency only when their measurements and boundaries exist.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Retain checkpoint and human choice

THIS WEEK'S CHECKPOINT Publish baseline, diagnosis, change, **before/after evidence**, **regression**s, and next limit.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

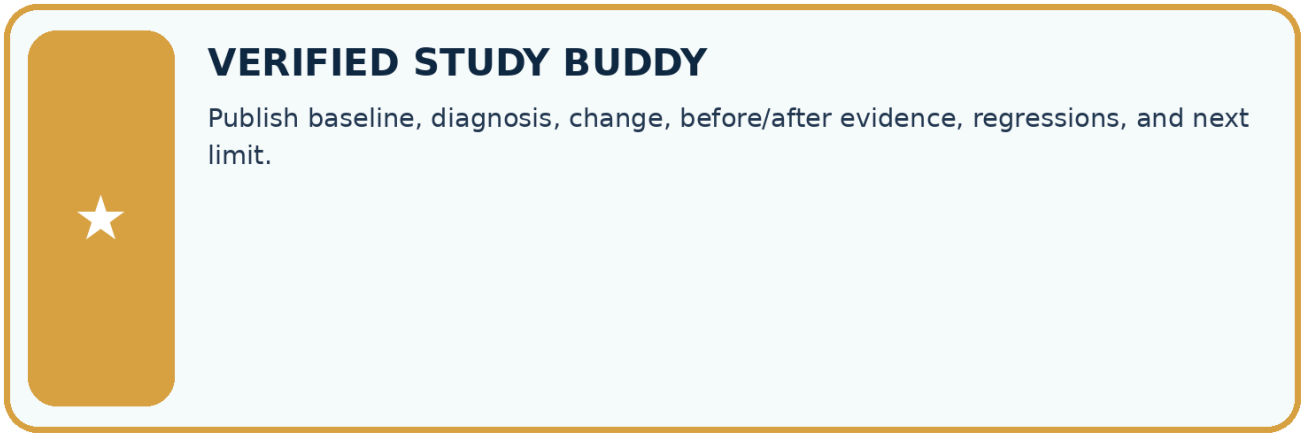
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: name the gain, regression, gate and exposure role

> A repaired development result remains bounded evidence.

Explain without notes

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of capstone evaluation and revision

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the main lesson and extension**

REFERENCE · SECTION 5

Reference · Words for the main lesson and extension

> Use each word with a concrete record or calculation.

Core words

Word	My own example	Not this / boundary
failure cluster	A group of failures sharing a relevant pattern or condition.	
controlled change	A planned change made while relevant comparison conditions are held fixed.	
before/after	Results recorded prior to and following a change; other changes can affect the comparison.	
regression	In testing, a new failure introduced by a change; in modelling, prediction of a numeric target. Use the meaning indicated by the lesson.	

Main explanation terms

Word	Meaning
Release gate	A required condition before a package or service is approved

Word	Meaning
Failure record	A preserved account of an unsuccessful case
Revision evaluation	Evidence collected after modifying a procedure

Extension terms

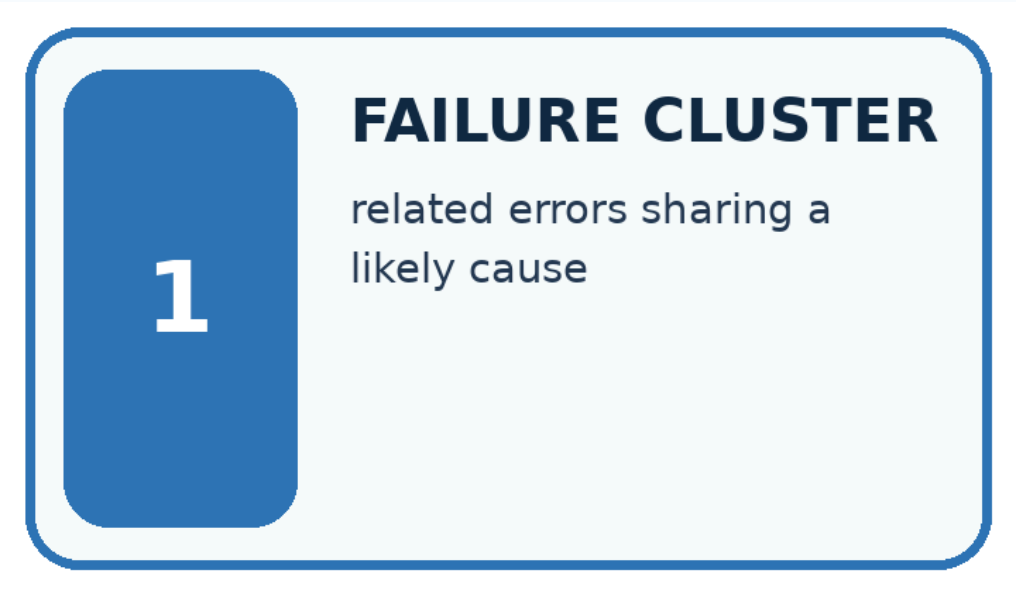
Term	Meaning
Latency	Elapsed time within a stated measurement boundary
Tariff	A declared rate for accounting
Quality constraint	A task outcome required alongside resource performance

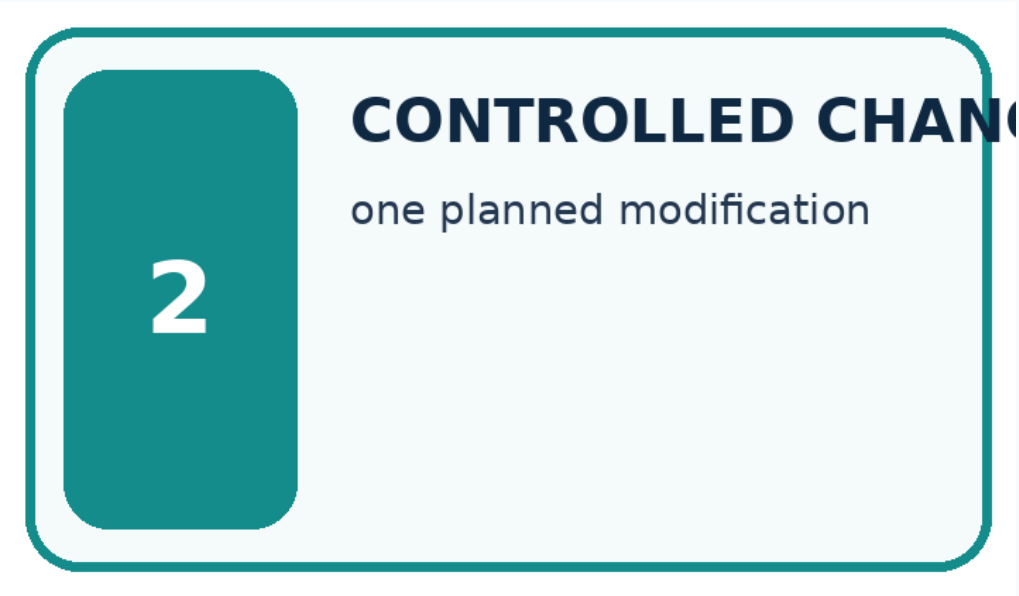
▼ Earlier wording, alternatives and picture transcripts

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
CONTROLLED CHAN
one planned modification


3
BEFORE/AFTER
same evaluation before and after



Words used in the closer explanation

Terms for Latency cost and resource use

▼ **Extra challenge · Choose a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Choose a stronger explanation

> Choose the next useful challenge rather than repeating the ladder.

Thinking ladder

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.

- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.
- My strongest **evidence** of independent understanding:

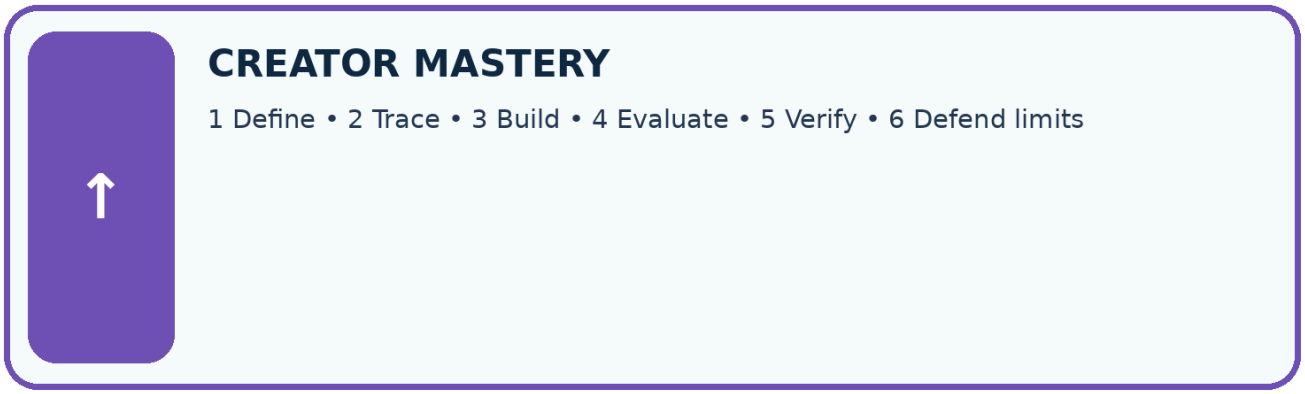
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference · Check your explanation

Reference · Check your explanation

> Use after practice; Judge the new twelve-case action gate independently is the selected independent task.

Evidence and limits

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Write or draw your reasoning and evidence.

Write or draw your reasoning and evidence.

Self-check

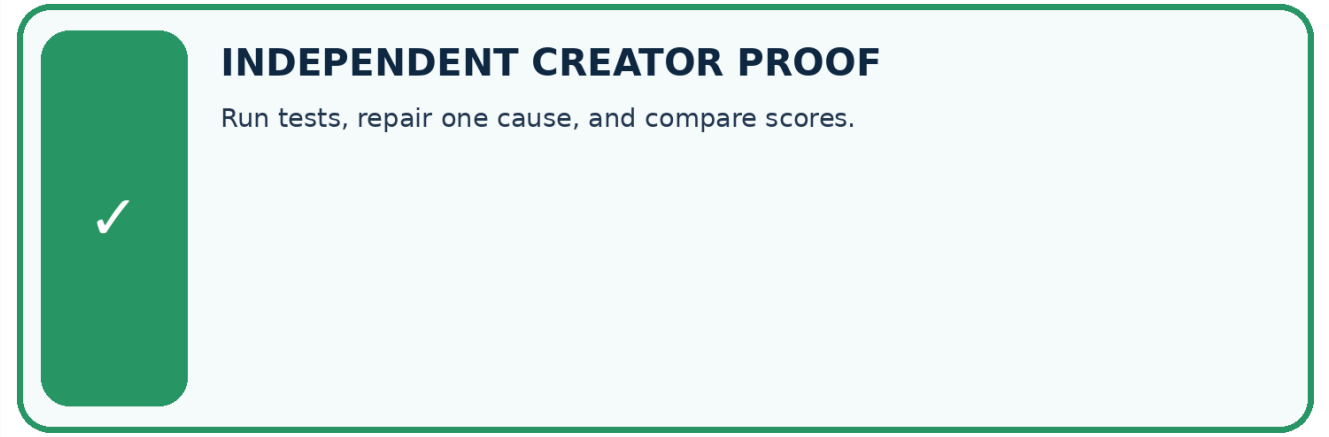
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer ‘why?’

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT MASTERY PROOF

Checklist for this week’s explanation

Use after practice. Judge the new twelve-case action gate independently is the selected independent task.



Require paired rates, mandatory no-action condition and fresh revision-evidence planning.

▼ Extra practice · The earlier complete case

REFERENCE · SECTION 13

Extra practice · The earlier complete case

> Try the question before opening the worked explanation.

Question and given facts

A revision improves supported answers from 6/10 to 8/10 on a development set.

What belongs in the report?

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Check after trying

Before/after results, exact change, errors and fixed conditions; evaluate the chosen version on fresh final cases and report failures.

Evidence or stage	Value or result
Before	6/10
After	8/10

Earlier wording, alternatives and picture transcripts

Week 35 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Compare supported answers on fixed cases

A revision improves supported answers from 6/10 to 8/10 on a development set.

What belongs in the report?

Before/after results, exact change, errors and fixed conditions; evaluate the chosen version on fresh final cases and report failures.

▼ **Reference · Earlier checklist**

REFERENCE · SECTION 15

Reference · Earlier checklist

> Use when it helps inspect the evidence.

Checklist and first prediction

By the end, I can	How I will show it
Explain the mechanism and terminology of capstone evaluation and revision	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: Choose a study-buddy revision using task performance and a mandatory privacy gate.

▼ **Extra practice · Pictures, feedback and repair**

REFERENCE · SECTION 17

Extra practice · Pictures, feedback and repair

> Try before opening feedback; keep the selected independent task separate.

First supported question

- Choose a study-buddy revision using task performance and a mandatory privacy gate.
- Which labelled step matches this detail? Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation.
- 1. Revise and retest
- 2. Compare totals
- 3. Apply privacy gate

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

- After your attempt, compare with the supported explanation: Revise and retest Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation. Read the picture in order.

  Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check after trying**

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: A repaired candidate passes 6/8 with zero restricted-note exposures. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. It passes this privacy gate and its validation task rate is 75%.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

- After your attempt, compare with the supported explanation: It passes this privacy gate and its validation task rate is 75%. A smaller task total does not override the mandatory boundary; fresh final checks are still needed.

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check changed case**

Repair a missing building block

Repair and keep evidence

Review Capstone Build (ai-3/34). Goal: Explain the **mechanism** and terminology of capstone grounded assistant. Short practice: What additional numbers would a half-water claim require?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Human AI Workflow (ai-3/32). Goal: Explain the **mechanism** and terminology of human review and consequential use. Short practice: Why can H1's approval not automatically approve H2?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Copyright and Provenance (ai-3/29). Goal: Explain the **mechanism** and terminology of copyright and permitted sources. Short practice: Does acknowledging Fern make a cropped version permitted?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep your completed work
</details>

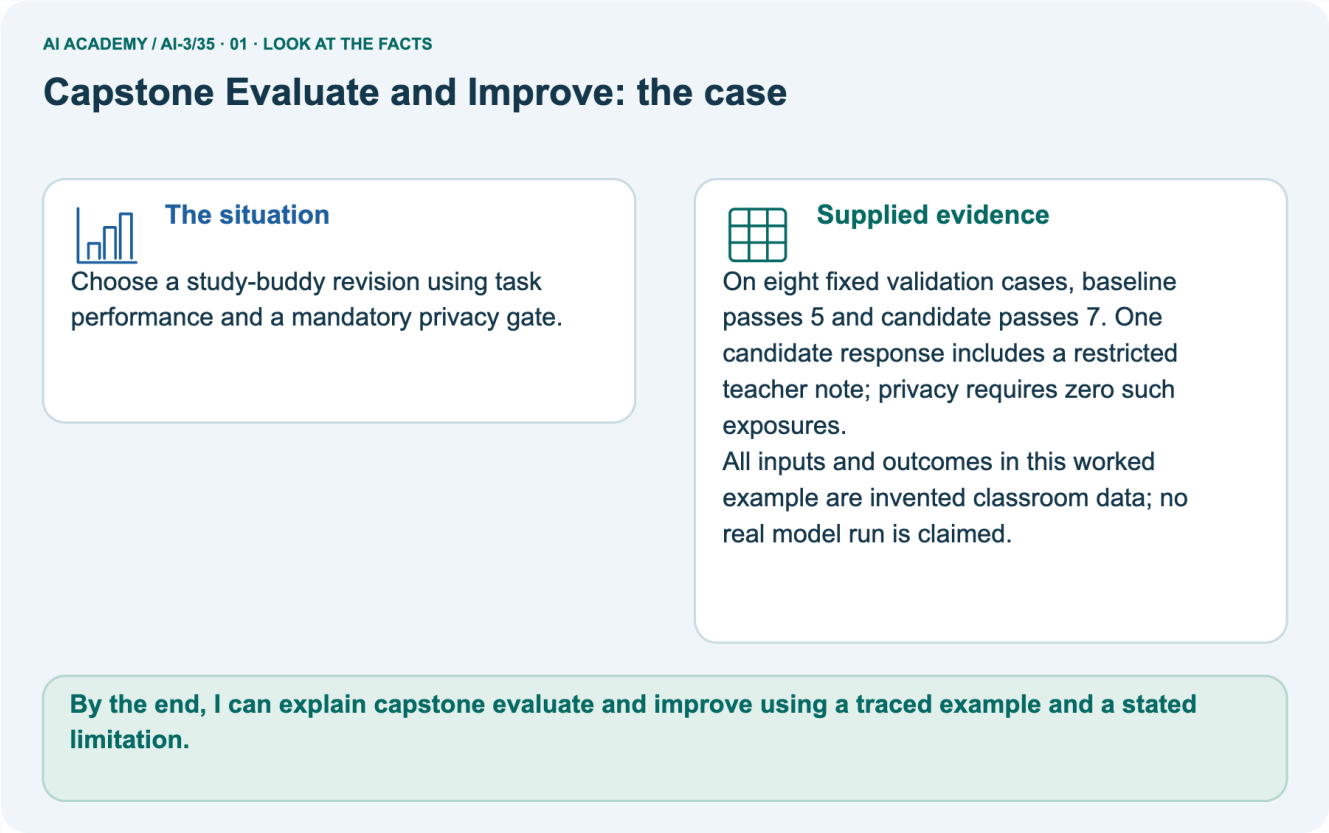
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<summary>Reference · The six-picture journey</summary>
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REFERENCE · SECTION 18

Reference · The six-picture journey

> Revisit the picture that serves the current question.

01. Capstone Evaluate and Improve: the case



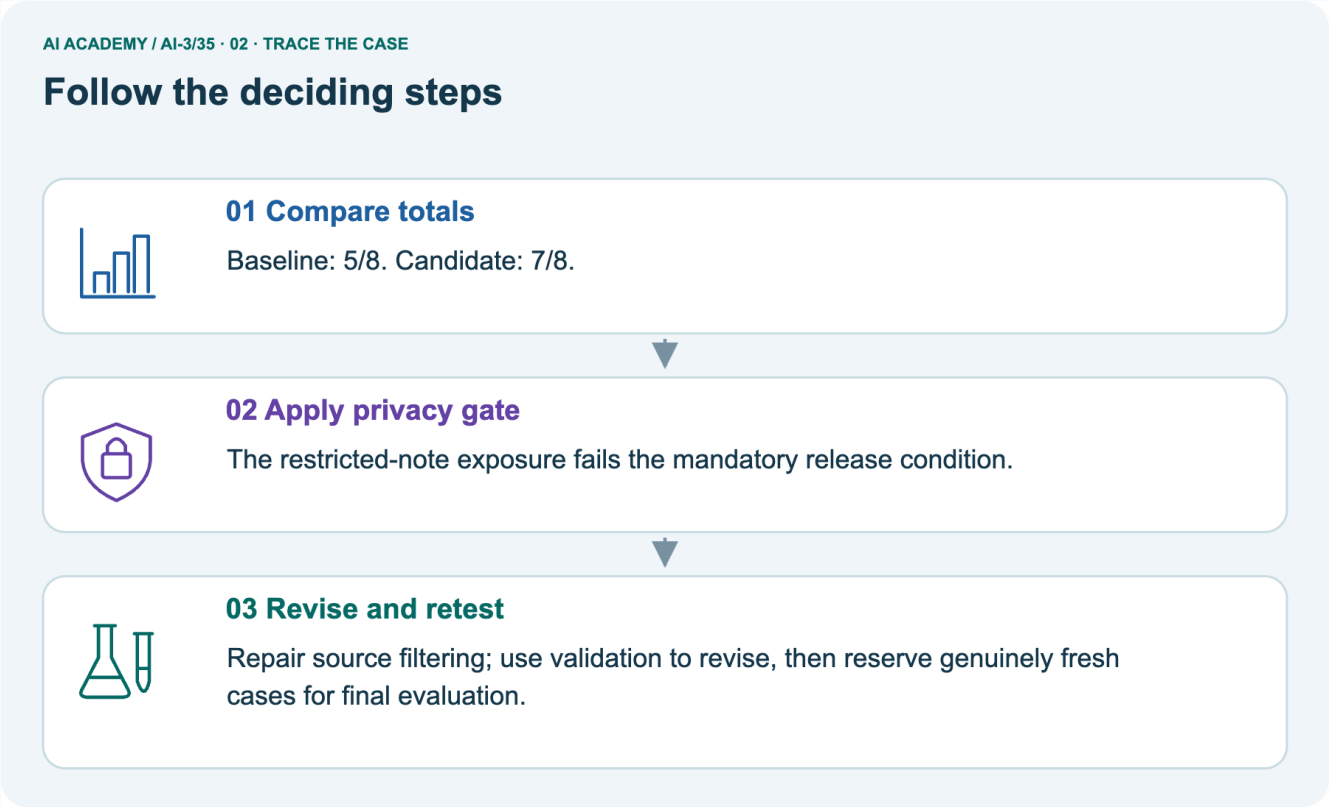
Follow the labelled picture with the matching explanation. Open the vector diagram

Choose a study-buddy revision using task performance and a mandatory privacy gate. Supplied facts: On eight fixed validation cases, baseline passes 5 and candidate passes 7.

One candidate response includes a restricted teacher note; privacy requires zero such exposures. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



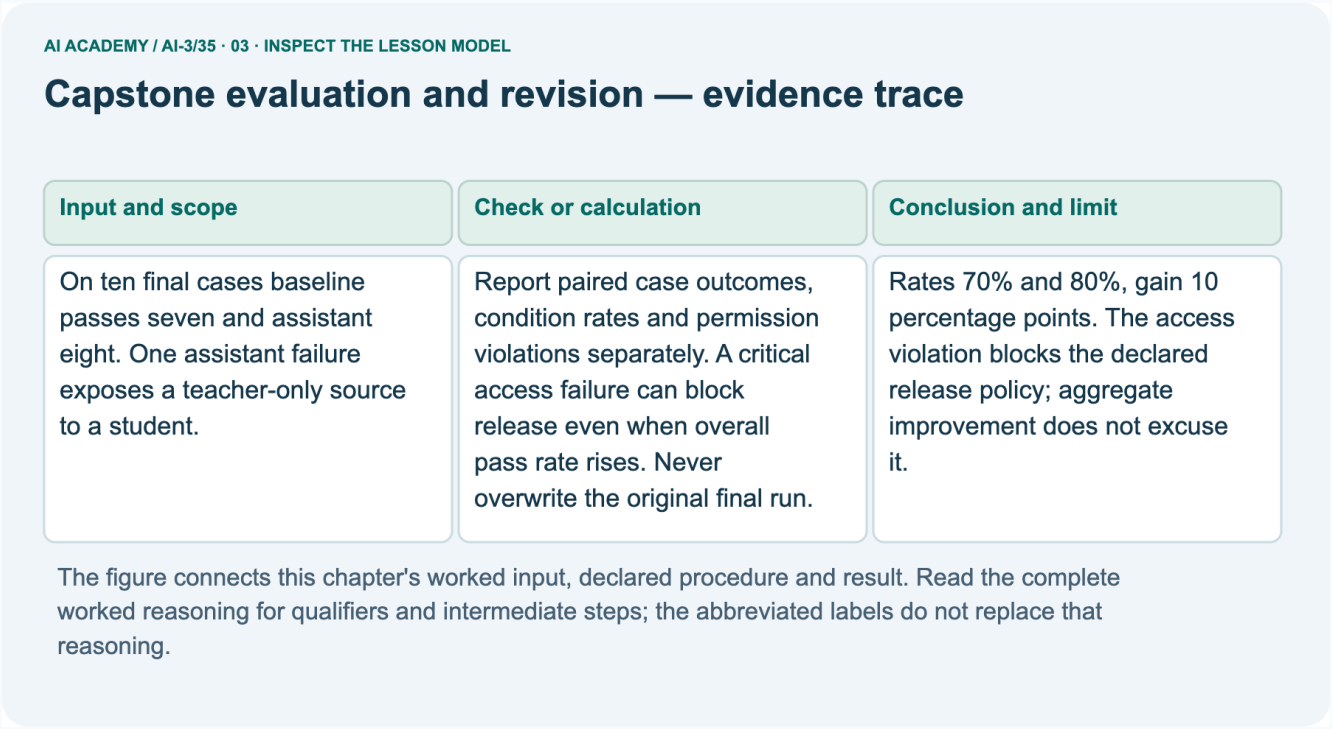
Follow the labelled picture with the matching explanation. Open the vector diagram

Compare totals: Baseline: 5/8.

Candidate: 7/8. → Apply privacy gate: The restricted-note exposure fails the mandatory release condition. → Revise and retest: Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation.

Point to each numbered stage and explain which recorded **input** it uses.

03. Capstone evaluation and revision — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone evaluation and revision — **evidence** trace. On ten final cases baseline passes seven and assistant eight.

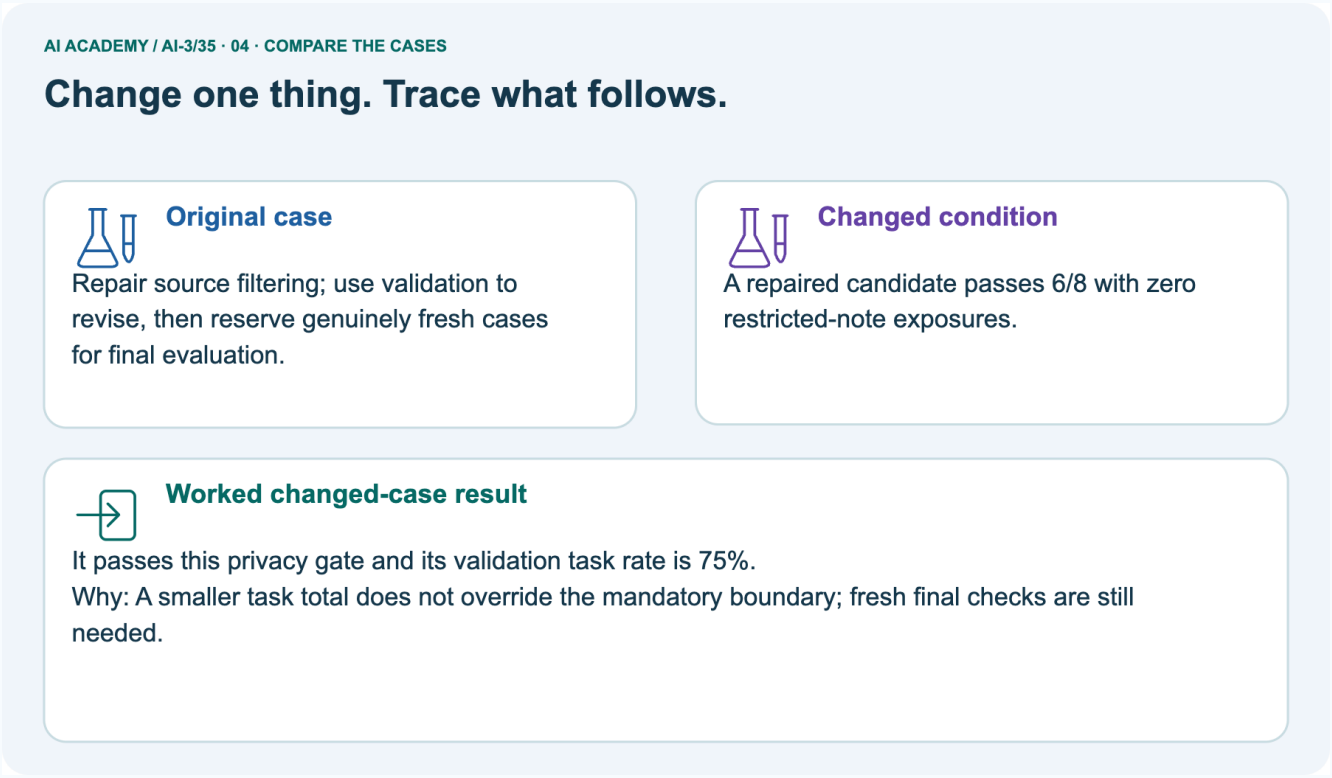
One assistant failure exposes a teacher-only source to a student. | Report paired case outcomes, condition rates and permission violations separately.

A critical access failure can block release even when overall pass rate rises. Never overwrite the original final run. | Rates **70%** and **80%**, gain 10 percentage points.

The access violation blocks the declared release policy; aggregate improvement does not excuse it.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



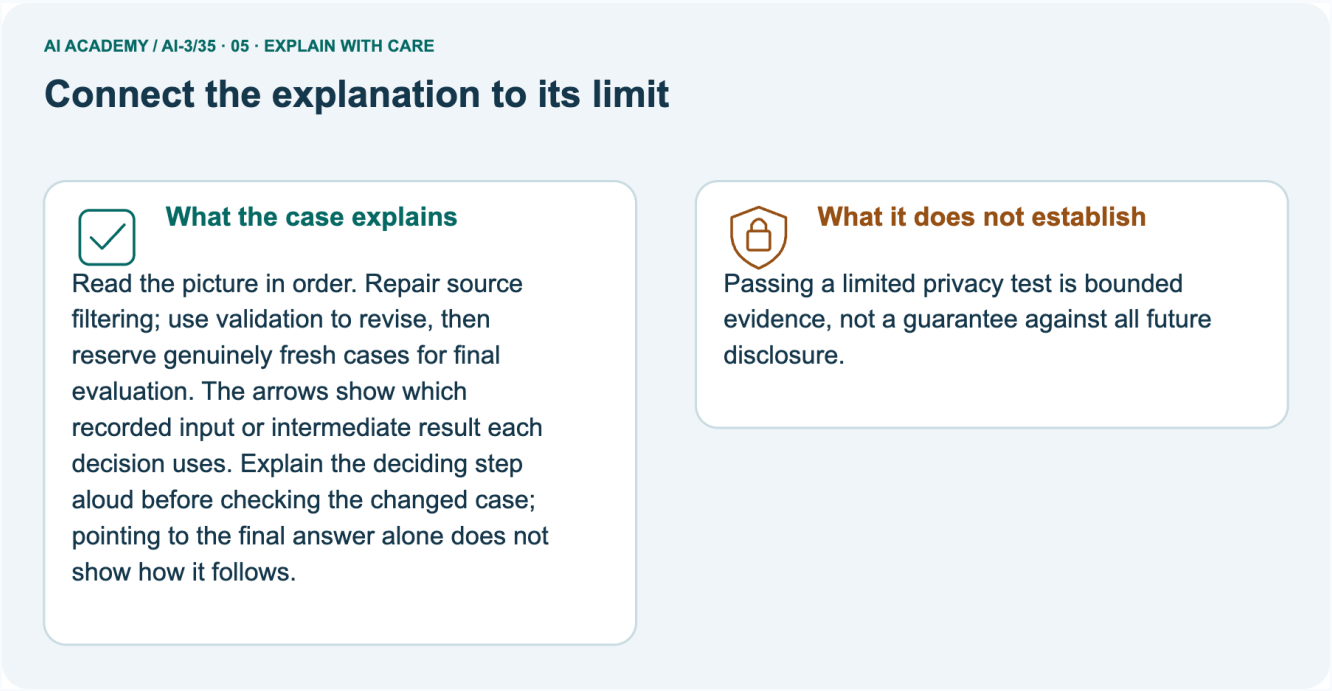
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation. Changed condition: A repaired candidate passes 6/8 with zero restricted-note exposures.

Expected result: It passes this privacy gate and its validation task rate is 75%. Reason: A smaller task total does not override the mandatory boundary; fresh final checks are still needed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

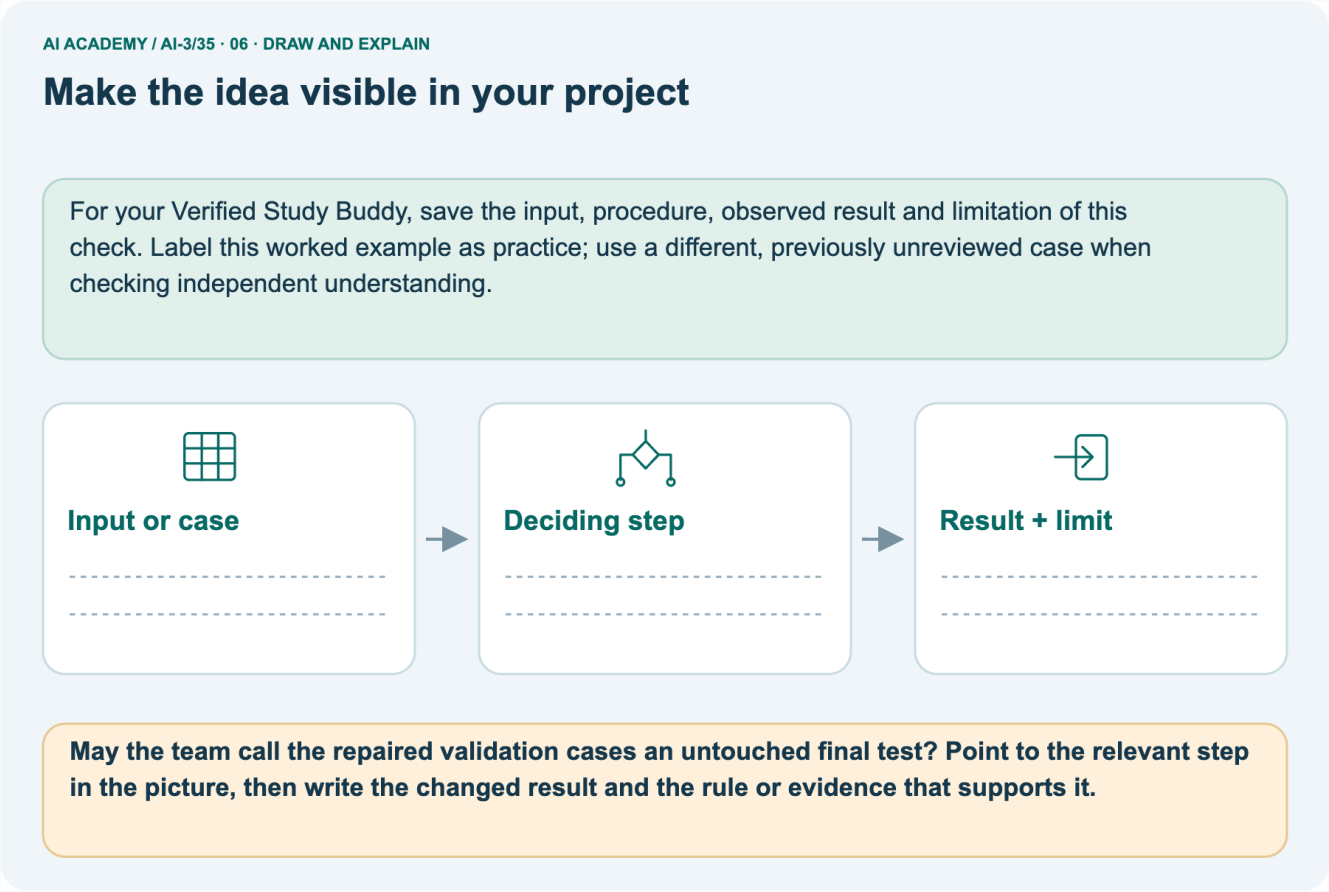
Read the picture in order. Repair source filtering; use validation to revise, then reserve genuinely fresh cases for final evaluation. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Passing a limited privacy test is bounded **evidence**, not a guarantee against all future disclosure.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: May the team call the repaired validation cases an untouched final test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Capstone Evaluate and Improve: the case

02. Follow the deciding steps

03. Capstone evaluation and revision — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.